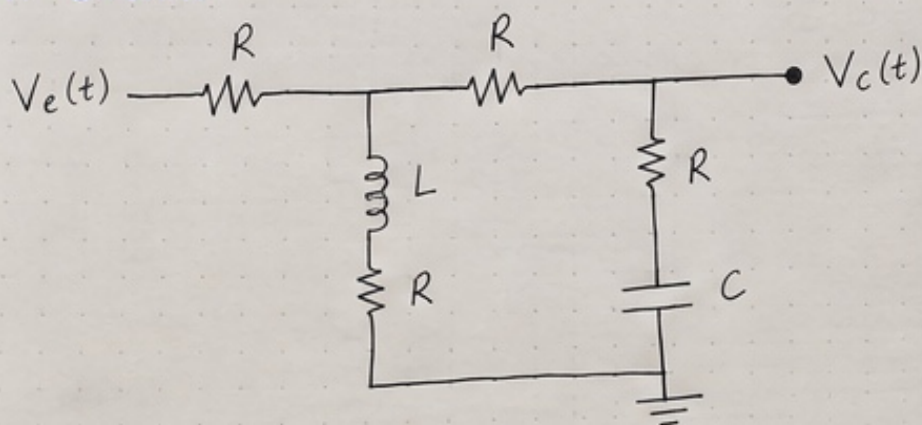


17 / sep / 2026

1. Circuito



2. Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R [i_2(t) - i_c(t)]$$

$$L \frac{d(i_1 - i_2(t))}{dt} + R [i_2 - i_c] = R i_2 + R i_c + \int i_2(t) dt$$

$$V_c(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

3. Transformada de la placa

$$V_c(s) = R i_1(s) + Ls [i_1(s) - i_2(s)] + R [i_2(s) - i_c(s)]$$

$$Ls [i_1 - i_2] + R [i_2 - i_c] = R i_2 + R i_c + \frac{i_2(s)}{Cs}$$

$$V_s(s) = R i_2(s) + \frac{i_2(s)}{Cs}$$

4. Procedimiento algebraico

- ① Agregar corrientes $i_1(s)$ e $i_2(s)$
- ② Expresar $V_c(s)$ en función de $i_2(s)$
- ③ Despejar $i_2(s)$ de la segunda ecuación

$$V_s(s) = \left(R + \frac{1}{Cs} \right) i_2(s) = \frac{CRs + 1}{Cs} i_2(s)$$

5. Simplificando

$$(Ls + R) \dot{i}_1(s) = \left(Ls + \frac{1}{Cs} - 3R \right) \dot{i}_2(s)$$

$$\dot{i}_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} - 3R \right) \dot{i}_2(s)$$

$$\dot{i}_1(s) = \frac{LCs^2 - 3RCs + 1}{Cs(Ls + R)} \dot{i}_2(s)$$

6. Sustituyendo en $V_c(s)$

$$V_c(s) = R \dot{i}_1(s) + Ls (\dot{i}_1(s) - \dot{i}_2(s)) + R \dot{i}_2(s) - R \dot{i}_2(s)$$

$$V_c(s) = [Ls + 2R] \left[\frac{LCs^2 + 3RCs + 1}{CLs^2 + CRs} \right] \dot{i}_2(s) - (Ls + R) \dot{i}_2(s)$$

$$V_c(s) = \left[\frac{(Ls + 2R)(LCs^2 + 3RCs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] \dot{i}_2(s)$$

$$V_c(s) = \left[\frac{CL^2s^3 + 5CLRs^2 + CL + 6CR^2s + 2R}{CL^2s^3 + 2CLR^2 + CR^2s} \right] \dot{i}_2(s)$$

$$V_c(s) = \left[\frac{3(CLRs^3 + 5CLR^2s + 2R)}{CLs^2 + CRs} \right] \dot{i}_2(s)$$

7. Función de Transferencia

$$\frac{V_s(s)}{V_c(s)} = \frac{\frac{CRs + 1}{Cs} \dot{i}_2(s)}{\left[\frac{3CLR^2s^3 + 5CLR^2s + 2R}{CLs^2 + CRs} \right] \dot{i}_2(s)}$$

$$\frac{V_s(s)}{V_c(s)} = \frac{(CRs + 1)(Ls + R)}{3CLR^2s^3 + (CL + 5CR^2)s + 2R}$$

8. Desarrollando FT

$$\frac{V_s(s)}{V_c(s)} = \frac{(CRs+1)(Ls+R)}{3CLR s^3 + (CL+5CR^2)s+2R}$$

9. Valores

$$L = 500 \mu H$$

$$C = 470 \mu F$$

$$R = 10 k\Omega$$

10. Error estacionario

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR+L)s + R}{3CLR s^3 + (CL+5CR^2)s + 2R} \right)$$

$$e(t) = 0.5 V$$

11. Estabilidad en lazo abierto

$$3CLR s^3 + (CL+5CR^2)s + 2R = 0$$

$$3(470 \times 10^{-6})(500 \times 10^{-6})(10,000) s^3 + (500 \times 10^{-6} + 5(470 \times 10^{-6})^2)s + 2(10,000) = 0$$

12. Raíces

$$X_1 = -0.085106$$

$$X_2 = -3.0803 \times 10^7$$

$$X_3 = -0.000006303$$

13. Respuesta al escalón

